9.5/10

Quiz 01 BSE-2A

Explain: int* const ptr and const int* arr with example (2) 1.5

int* const ptr prevents from updating the memory address that the pointer points to while const int* arr prevents from updating the value of the variable it points to.

```
int x = 10;
int *const ptr = &x;
int y = 5;
ptr = &y; // invalid
*ptr = 20; // valid
```

example of const int*?

```
int process(const int*arr, int start, int end, int add = 1){
    int sum = 0;
    const int*p = arr + start;
    for (int i = start; i<end; i++){
        sum+= *p+add;
        p++
    } return sum;
}
```

Dry Run the code step by step and show output(8)

```
#include <iostream>
using namespace std;

int process(const int* arr, int start, int end, int add = 1) {
    int sum = 0;
    const int* p = arr + start;

    for (int i = start; i < end; i++) {
        sum += *p + add;
        p++;
    }
    return sum;
}

int main() {
    int a[5] = { 2, 4, 6, 8, 10 };

    int* const ptr = a;

    int r1 = process(ptr, 1, 4);
    *(a+2) = 1;
    int r2 = process(ptr, 0, 3, 0);

    cout << r1 << endl;
    cout << r2 << endl;

    return 0;
}
```

value not be modified

first index

start end

8

2	4	6	8	10

ptr

r1 :-
start = 1, end = 4
sum = 0
*p = 2+1; // *p=3
① i=1 ; i<4; i++
sum = sum + *p + add = 5
② sum = 12
③ sum = 21

r2 :-
① sum = 6
② sum = 7

Output :-
21
7